

ASSOCIATIVE SUBMANIFOLDS IN JOYCE–KARIGIANNIS MANIFOLDS

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ABSTRACT. Joyce and Karigiannis constructed torsion-free G_2 -structures on resolutions of G_2 -orbifolds whose singularities are modelled on $\mathbb{R}^3 \times (\mathbb{R}^4/\{\pm 1\})$, by gluing in families of Eguchi–Hanson spaces parametrised by a nowhere vanishing harmonic 1-form λ on the singular set L . Using the control of the resulting torsion-free G_2 -structures obtained recently by Majewski, we construct associative submanifolds in these G_2 -manifolds. They are diffeomorphic to $S^1 \times S^2$, they are invariant under a finite group of symmetries, and, as the gluing parameter tends to zero, they collapse onto closed geodesics of L and their volumes tend to zero. This extends part of the construction of associative submanifolds in Joyce’s generalised Kummer constructions by Dwivedi, Platt and Walpuski. As an example we exhibit 160 such associative submanifolds in a G_2 -manifold with $b^2 = 4$ and $b^3 = 67$ obtained from $T^3 \times K3$.

1. INTRODUCTION

In [DPW23] associative submanifolds were constructed in the G_2 -manifolds arising from Joyce’s generalised Kummer construction [Joy96a, Joy96b, Joy00], i.e. from resolutions of flat G_2 -orbifolds T^7/Γ . As the G_2 -manifolds degenerate to the orbifold, the volumes of these associative submanifolds tend to zero, as predicted by Halverson and Morrison [HM16, §6.2]. Joyce and Karigiannis [JK21] have extended the generalised Kummer construction to compact torsion-free G_2 -orbifolds (M', φ, g) which are not necessarily flat, whose singular set L is a compact associative 3-fold with transverse singularity model $\mathbb{R}^4/\{\pm 1\}$, provided L carries a closed, coclosed, nowhere vanishing 1-form λ . The singularities are resolved by gluing in a bundle $\sigma: P \rightarrow L$ of Eguchi–Hanson spaces whose bolts have area $\pi t^2 |\lambda|$, see [JK21, Theorem 1.1]. Already in [DPW23, 4.10] it was pointed out that the methods of [DPW23] ought to produce associative submanifolds in the resulting G_2 -manifolds.

The missing ingredient was analytic. The perturbation argument of [DPW23, §3] requires control of the difference between the torsion-free G_2 -structure φ_t and the local model φ_t^P near the exceptional set in $C^{1,\alpha}$ with respect to the rescaled metric $t^{-2}g_t^P$, with a positive power of t (cf. [DPW23, 2.18, 2.19]). In the flat case this control is provided by [Pla25, Theorem 4.29]. For the Joyce–Karigiannis manifolds, however, the existence proof in [JK21] goes through Joyce’s existence theorem [Joy00, Theorem 11.6.1] and yields only the C^0 -estimate $\|\tilde{\varphi}_t^N - \varphi_t^N\|_{C^0} \leq K_4 t^{1/18}$, see the proof of [JK21, Theorem 6.4]. Recently, Majewski [Maj26], building on the uniform elliptic theory of [Maj25], has given a different existence proof for torsion-free $\text{Spin}(7)$ - and G_2 -structures on resolutions of orbifolds, which comes with estimates in weighted Hölder norms adapted to the gluing geometry [Maj26, Theorem 6.1, Corollary 6.2]. The purpose of this article is to explain how these estimates yield the required control (Theorem 2.12) and to carry out the construction of associative submanifolds in Joyce–Karigiannis manifolds. Since Theorem 2.12 gives control of order t^θ with $\theta < 1$ only, we cannot perturb Morse–Bott families of associative submanifolds as in [DPW23, Proposition 3.18]; instead we use the equivariant, rigid version of the argument sketched in [DPW23, 4.2], see Remark 1.4.

Throughout, (M', φ, g) , L , λ are as in [JK21, §6.5–§6.6]; that is, (M', φ, g) is a compact torsion-free G_2 -orbifold whose singular set is a compact 3-dimensional submanifold L near which M' is modelled on $\mathbb{R}^3 \times (\mathbb{R}^4/\{\pm 1\})$, and λ is a closed, coclosed, nowhere vanishing 1-form on $(L, g|_L)$, possibly twisted by a principal $\mathbb{Z}_2$ -bundle $Z \rightarrow L$; see Definition 2.1. We denote by $\pi: N \rightarrow M'$ the resolution of [JK21, §6.1] and by $Q = \pi^{-1}(L) \subset N$ the exceptional

set; $\Pi = \pi|_Q: Q \rightarrow L$ is a bundle with fibre S^2 . For $t > 0$ small, [Theorem 2.12](#) provides torsion-free G_2 -structures φ_t on N which are close to the fibred model (φ_t^P, g_t^P) of [\[JK21, \(4.23\)\]](#) near Q .

Theorem 1.1. *Let $(M', \varphi, g, L, \lambda)$ be as above and let $(\varphi_t)_{t \in (0, t_0)}$ be the torsion-free G_2 -structures on N of [Theorem 2.12](#), with $\alpha \in (0, 1)$ and $\theta > 0$ as there. Let K be a finite group acting on M' by automorphisms of (φ, g) preserving L and $\pm\lambda$, and let K act on N by the induced action; then φ_t is K -invariant. Let L_0 be a connected component of L and suppose that:*

- (a) *There are $\ell > 0$, a 6-dimensional orbifold W with a torsion-free $SU(3)$ -structure (ω, Ω) whose singular set is a closed complex curve $\Sigma \subset W$ with transverse singularity model $\mathbb{C}^2/\{\pm 1\}$, a K -invariant open neighbourhood U of L_0 in M' and an open neighbourhood U_W of Σ in W , together with an isomorphism of G_2 -orbifolds*

$$U \cong (\mathbb{R}/\ell\mathbb{Z}) \times U_W, \quad \varphi = dx \wedge \omega + \operatorname{Re} \Omega, \quad (1.2)$$

which maps L_0 to $(\mathbb{R}/\ell\mathbb{Z}) \times \Sigma$ and $\lambda|_{L_0}$ to $\pm dx$, and under which K acts by $\operatorname{id} \times K_W$ for a group K_W of automorphisms of (U_W, ω, Ω) .

- (b) *$c \in \Sigma$ is a fixed point of K_W and $T_c\Sigma$ contains no non-zero K_W -invariant vector.*

Set $Q_c := \Pi^{-1}((\mathbb{R}/\ell\mathbb{Z}) \times \{c\}) \subset Q$. Then there are $t_1 \in (0, t_0]$, $C > 0$ and $r_0 > 0$ with the following significance. For every $t \in (0, t_1)$ there is a K -invariant section v_t of the normal bundle of Q_c in $(N, t^{-2}g_t^P)$ with $\|v_t\|_{C^{1,\alpha}} \leq Ct^\theta$, such that the normal graph P_t of v_t is a φ_t -associative submanifold of N ; moreover P_t is the only such graph of a K -invariant section of $C^{1,\alpha}$ -norm at most r_0 . The submanifold P_t is diffeomorphic to $S^1 \times S^2$, it represents the homology class $[Q_c] \in H_3(N; \mathbb{Z})$, its volume with respect to g_{φ_t} is $\pi\ell t^2(1 + O(t^\theta))$, and it collapses onto the closed geodesic $(\mathbb{R}/\ell\mathbb{Z}) \times \{c\} \subset L_0 \subset M'$ as $t \rightarrow 0$.

Here and throughout, norms are computed with respect to the rescaled metric $\tilde{g}_t := t^{-2}g_t^P$ unless said otherwise; cf. [\[DPW23, 2.18\]](#). In [\(1.2\)](#), x denotes the coordinate on $\mathbb{R}/\ell\mathbb{Z}$; the singular set of $U \cong (\mathbb{R}/\ell\mathbb{Z}) \times U_W$ is then $(\mathbb{R}/\ell\mathbb{Z}) \times \Sigma$, whose transverse singularity model is $\mathbb{R}^4/\{\pm 1\}$, as required. Distinct points $c \neq c'$ satisfying (b) (possibly for different groups K) give distinct associative submanifolds once t is small, since P_t lies within g_t^P -distance $Ct^{1+\theta}$ of Q_c , whereas $\sigma: (P, g_t^P) \rightarrow (L, g|_L)$ does not increase distances and maps Q_c onto $(\mathbb{R}/\ell\mathbb{Z}) \times \{c\}$.

The situation of [Theorem 1.1](#) arises for the G_2 -orbifolds $(T^3 \times X)/\Gamma$ of [\[JK21, §7.3\]](#), with X a hyperkähler $K3$ surface. We work out [\[JK21, Example 7.2\]](#) in detail in [Section 4](#).

Corollary 1.3. *Let (N, φ_t) be the G_2 -manifolds with $b^2(N) = 4$, $b^3(N) = 67$ and $\operatorname{Hol}(g_{\varphi_t}) = G_2$ obtained in [\[JK21, Example 7.2\]](#) by resolving $M' = (T^3 \times X)/\mathbb{Z}_2^2$, where $X \rightarrow \mathbb{CP}^2$ is the double cover branched along the Fermat sextic $C = \{z_0^6 + z_1^6 + z_2^6 = 0\}$, equipped with the torsion-free G_2 -structures of [Theorem 2.12](#). The singular set of M' consists of two copies of $S^1 \times C$ and two copies of $S^1 \times S^2$. For all sufficiently small $t > 0$, (N, φ_t) contains at least 160 pairwise distinct associative submanifolds diffeomorphic to $S^1 \times S^2$, namely 54 collapsing onto closed geodesics in each of the components $S^1 \times C$ and 26 collapsing onto closed geodesics in each of the components $S^1 \times S^2$. Their volumes are $\pi t^2(1 + O(t^\theta))$, and the 54 (resp. 26) associative submanifolds belonging to one component of the singular set are homologous to each other and homologically non-trivial.*

Remark 1.4. In [\[DPW23\]](#) the associative submanifolds are obtained by perturbing a Morse–Bott family $\mathbf{P}_0: M \rightarrow \mathcal{S}$ of associative submanifolds of the model G_2 -structure, using [\[DPW23, Proposition 3.18\]](#), and then finding zeros of a 1-form on the parameter space M by topological arguments [\[DPW23, 3.6\]](#). In the situation of [Theorem 1.1](#), the submanifolds Q_c , $c \in \Sigma$, form such a Morse–Bott family (with $M = \Sigma$, cf. [\[DPW23, 3.12, 3.16\]](#) and [Lemma 3.7](#) below). However, the contraction argument in the proof of [\[DPW23, Proposition 3.18\]](#) requires $t^{\beta-2\gamma} \rightarrow 0$, i.e. $\beta > 2\gamma$, where $t^{-\gamma}$ is the loss in the inverse of the linearised operator caused by the length $\sim t^{-1}$ of the S^1 -factor of the model associatives ([\[DPW23, Proposition 3.22\]](#); $\gamma = 1$) and t^β is the $C^{1,\alpha}$ -distance of the torsion-free G_2 -structure to the model. In [\[DPW23, §4\]](#), $\beta = 5/2$ by

[Pla25, Theorem 4.29], whereas Theorem 2.12 only gives $\beta = \theta < 1$ (Remark 2.14). This is why we restrict to K -invariant deformations, for which the loss $t^{-\gamma}$ disappears (Lemma 3.8) and any $\theta > 0$ suffices; this is the “simpler proof” alluded to in [DPW23, 4.2]. Should the control in Theorem 2.12 be improved to some $\theta > 2$, [DPW23, Proposition 3.18] would apply verbatim to the family $(Q_c)_{c \in \Sigma}$, and the Poincaré–Hopf argument of [DPW23, 3.6 (1)] would produce associative submanifolds near $Q|_{L_0}$ whenever $\chi(\Sigma) \neq 0$, without any symmetry assumption (at least $|\chi(\Sigma)|$ of them counted with multiplicity, and at least two if $\Sigma \cong S^2$ by [DPW23, 3.6 (3)]).

Remark 1.5. The associative submanifolds of Theorem 1.1 are “thin”: in the metric g_{φ_t} they look like the product of a circle of length ℓ with a round 2-sphere of area πt^2 . As in [DPW23], they exhibit degenerating families of G_2 -manifolds carrying associative submanifolds whose volumes tend to zero; cf. [DPW23, §1, (3)] and [HM16, §6.2].

Conventions. We follow the sign conventions of [JK21], so that on $\mathbb{R}^3 \times X$, for a hyperkähler 4-manifold $(X, h, \omega^I, \omega^J, \omega^K)$,

$$\varphi = dx_1 \wedge dx_2 \wedge dx_3 - dx_1 \wedge \omega^I - dx_2 \wedge \omega^J - dx_3 \wedge \omega^K, \quad g = dx_1^2 + dx_2^2 + dx_3^2 + h \quad (1.6)$$

is a torsion-free G_2 -structure [JK21, (2.13)]. The cross product is defined by $g(u \times v, w) = \varphi(u, v, w)$. A closed oriented 3-dimensional submanifold P of a 7-manifold with a G_2 -structure φ is *associative* if $\varphi|_P = \text{vol}_P$; if $d\varphi = 0$ this is equivalent to [DPW23, Definition 3.1] by [HL82, Theorem IV.1.6], and in general we take [DPW23, Definition 3.1] as the definition. For a submanifold P and a G_2 -structure φ we write $D_P^\varphi = -\gamma \nabla + \tau^\perp$ for the Fueter operator of [DPW23, Definition 3.8].

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2. THE JOYCE–KARIGIANNIS CONSTRUCTION AND MAJEWSKI’S ESTIMATE

2.1. Joyce–Karigiannis data and the resolution.

Definition 2.1. *Joyce–Karigiannis data* consist of the following (see [JK21, Theorem 6.4, §6.5, §6.6]):

- (i) a compact torsion-free G_2 -orbifold (M', φ, g) whose singular set L is a compact 3-dimensional submanifold such that M' is locally modelled on $\mathbb{R}^3 \times (\mathbb{R}^4 / \{\pm 1\})$ near L ; L is totally geodesic and associative, cf. [JK21, Proposition 2.13];
- (ii) a principal $\mathbb{Z}_2$ -bundle $Z \rightarrow L$ and a closed, coclosed, nowhere vanishing Z -twisted 1-form λ on $(L, g|_L)$, i.e. a section of $T^*L \otimes_{\mathbb{Z}_2} Z$ with $d\lambda = d^*\lambda = 0$ [JK21, §6.6]. (If Z is trivial, λ is an ordinary 1-form as in [JK21, Theorem 6.4].)

The case $M' = M / \langle \iota \rangle$ for a G_2 -involution ι of a compact torsion-free G_2 -manifold M [JK21, Definition 2.11] is the one treated in the main text of [JK21]; the general case requires no changes [JK21, §6.5]. We allow L to be disconnected; the components are treated independently.

Fix Joyce–Karigiannis data. Following [JK21, §3.1, §4.1], let $\nu \rightarrow L$ be the normal bundle of L (a rank 4 real vector bundle with fibre metric h_ν and connection ∇^ν induced from g), let $r: \nu \rightarrow [0, \infty)$ be the radius function, $U_R := \{r < R\}$, and let $\Upsilon: U_R \rightarrow M'$ be a tubular neighbourhood map as in [JK21, Definition 3.2]; we take Υ to be the normal exponential map of g , so that Υ is compatible with all isometries of (M', g) preserving L . Write $A := |\lambda|_{g_L}: L \rightarrow (0, \infty)$ and $e_1 := A^{-1}\lambda$, a unit (Z -twisted) 1-form. Cross product with the vector field dual to e_1 defines a complex structure $I = e_1 \times (\cdot)$ on the fibres of ν [JK21, Remark 4.1], which is defined up to sign in the twisted case. Let $\rho: P \rightarrow \nu / \{\pm 1\}$ be the fibrewise blow-up of $\nu / \{\pm 1\}$ along the zero section with respect to I [JK21, §4.1] (it does not depend on the sign of I [JK21, §6.6]); $\sigma: P \rightarrow L$ is a bundle with fibre the

Eguchi–Hanson space X , $Q := \rho^{-1}(0(L))$ is a bundle $\Pi := \sigma|_Q: Q \rightarrow L$ with fibre S^2 , and $\tilde{r} := r \circ \rho: P \rightarrow [0, \infty)$ [JK21, Remark 4.3]. For $t > 0$ and $R > 0$ we write

$$P^R := \{\tilde{r} \leq R\} \subset P. \quad (2.2)$$

For $t > 0$, the resolution N is obtained by gluing $\rho^{-1}(U_{t^{-1}R}/\{\pm 1\}) \subset P$ to $M' \setminus L$ along $\Upsilon_t \circ \rho$, where $\Upsilon_t(x, a) := \Upsilon(x, ta)$ [JK21, (6.1)]; it comes with a continuous map $\pi: N \rightarrow M'$ restricting to a diffeomorphism $N \setminus Q \rightarrow M' \setminus L$. We regard P^R , for fixed R and t small, as a subset of N . By [JK21, Proposition 6.1, (6.32)],

$$b^k(N) = b^k(M') + b^{k-2}(L; Z) \quad \text{and} \quad \pi_1(N) \cong \pi_1(M'), \quad (2.3)$$

where $b^k(L; Z)$ denotes the Z -twisted Betti numbers; if Z is trivial then $Q \cong L \times S^2$ [JK21, proof of Proposition 6.1].

2.2. The fibred model. Let $\check{\nabla}$ be a connection on ν compatible with h_ν and I , and let $TP = \check{V} \oplus \check{H}$ be the induced splitting into vertical and horizontal subbundles; see [JK21, §4.2], where such a connection is used near Q .¹ This gives forms of type (i, j) on P (i vertical and j horizontal degrees) [JK21, (4.5)]. As in [JK21, §4.3–§4.4] (using a local oriented orthonormal coframe (e_1, e_2, e_3) of L , in the twisted case defined up to sign), define the 2-forms $\check{\omega}^I, \check{\omega}^J, \check{\omega}^K$ of type $(2, 0)$ and the fibre metric $\check{g}_{2,0}$ on P by [JK21, (4.7), (4.14), §4.4], and set

$$\begin{aligned} \varphi_t^P &:= \sigma^*(\text{vol}_L) - t^2(\sigma^*(e_1) \wedge \check{\omega}^I + \sigma^*(e_2) \wedge \check{\omega}^J + \sigma^*(e_3) \wedge \check{\omega}^K), \\ g_t^P &:= \sigma^*(g_L) + t^2 \check{g}_{2,0}, \end{aligned} \quad (2.4)$$

which is the G_2 -structure of [JK21, (4.21), (4.23)], with $\psi_t^P = *_{g_t^P} \varphi_t^P = t^4 \check{\psi}_{4,0} + t^2 \check{\psi}_{2,2}$ as there. We recall the relevant properties [JK21, §2.5, §4.3, §4.4]: for $x \in L$ the fibre $(P_x, \check{g}_{2,0}, \check{\omega}^I, \check{\omega}^J, \check{\omega}^K)$ is isomorphic to the Eguchi–Hanson space $(X, h_{A(x)}, \omega_{A(x)}^I, \omega^J, \omega^K)$ of [JK21, (2.16)–(2.21)]; the bolt $Q_x \subset P_x$ is I -holomorphic and has area $\pi A(x)$ with respect to $\check{g}_{2,0}$ [JK21, Remark 2.10]; and $\check{\omega}^J, \check{\omega}^K$ vanish on Q , because $\omega^J + i\omega^K$ is the pull-back of $dz_1 \wedge dz_2$ under the blow-down map, which contracts the bolt [JK21, §2.5]. Finally we use the rescaled metric and G_2 -structure

$$\tilde{g}_t := t^{-2} g_t^P = t^{-2} \sigma^*(g_L) + \check{g}_{2,0}, \quad \tilde{\varphi}_t^P := t^{-3} \varphi_t^P, \quad (2.5)$$

cf. [DPW23, 2.18]: on P^R the metric $\tilde{g}_t$ has curvature bounded and injectivity radius bounded below uniformly in t ; its fibres are Eguchi–Hanson spaces of size one and its base L has been blown up by the factor t^{-1} . All Hölder norms on P^R are taken with respect to $\tilde{g}_t$ unless said otherwise. Note that for a k -form Ψ we have $|t^{-k} \Psi|_{\tilde{g}_t} = |\Psi|_{g_t^P}$.

2.3. Majewski’s theorem. We now describe the results of [Maj26] in the present setting. In [Maj26, §6.2.1] a G_2 -orbifold (M', φ) is resolved by resolving the $\text{Spin}(7)$ -orbifold

$$(\mathbb{T} \times M', \Phi := ds \wedge \varphi + *_g \varphi), \quad \mathbb{T} := \mathbb{R}/\mathbb{Z}, \quad (2.6)$$

$\mathbb{T}$ -equivariantly. Its singular stratum is $S := \mathbb{T} \times L$, of codimension four with isotropy group $\Gamma = \{\pm 1\} \subset \text{Sp}(1)$; this is case (2) of [Maj26, §2.2], and the stratum is of type A in the sense of [Maj26, Appendix A.1]: the isotropy group $\{\pm 1\}$ is central in $\text{SO}(4)$, so the link bundle is a smooth $\mathbb{RP}^3$ -bundle, cf. [Maj26, §3.2.1]. The normal cone bundle is $N_0 = \mathbb{T} \times \nu/\{\pm 1\}$ [Maj26, §3.2.1]. Resolution data for such a stratum are sections of the parameter bundle $\mathfrak{P}_\Gamma = \Lambda_+^2 T^* S \otimes \mathfrak{H}_\Gamma^2$, where the McKay local system $\mathfrak{H}_\Gamma^2$ is a flat real line bundle for $\Gamma = \{\pm 1\}$ [Maj26, §5.3.2]. Given λ , set

$$\zeta := ds \wedge \lambda + *_L \lambda \in \Omega_+^2(S, \mathfrak{H}_\Gamma^2), \quad (2.7)$$

where $\mathfrak{H}_\Gamma^2 \cong Z \times_{\mathbb{Z}_2} \mathbb{R}$; here we normalise the identification $\Lambda_+^2 T_{(s,x)}^* S \otimes \mathfrak{H}_\Gamma^2 \cong \text{Im } \mathbb{H} \otimes \mathfrak{h}_\Gamma$ of [Maj26, §5.3.2] so that the Eguchi–Hanson space assigned to $\zeta(s, x)$ by Kronheimer’s construction

¹In [JK21, §4.2] the splitting is interpolated with the one induced by the connection $\check{\nabla}^\nu$ of [JK21, §3.4] for $\tilde{r} \geq 1$, which is needed for the torsion estimates of [JK21] but not here. If $\nabla^\nu I = 0$, as in Lemma 3.3 below, one may take $\check{\nabla} = \nabla^\nu$.

[Kro89] is the one with parameter $A(x) = |\lambda_x|$, i.e. with bolt of area $\pi A(x)$, and with complex structure $I = e_1 \times (\cdot)$ (any other normalisation amounts to replacing λ by a constant multiple). Since $d\zeta = -ds \wedge d\lambda + d*_L\lambda$, the section ζ is closed and coclosed if and only if λ is, and ζ is *regular* (avoids the walls of [Maj26, §5.3.2], which for $\Gamma = \{\pm 1\}$ is the zero section) if and only if λ is nowhere vanishing. Hence, by [Maj26, Theorem 5.1], ζ determines an *adiabatic torsion-free* ACF Spin(7)-resolution $\rho_\zeta: (N_\zeta, \Phi_\zeta) \dashrightarrow (N_0, \Phi_0)$ of the normal cone bundle [Maj26, Proposition 5.1, Proposition 5.2, Theorem 5.1], defined as the pull-back of the universal family $\mathfrak{N}_\Gamma \rightarrow \mathfrak{P}_\Gamma$ with its universal Cayley form [Maj26, §5.3.2]. By construction and the naturality of the universal family, $N_\zeta = \mathbb{T} \times P$ with $\mathbb{T}$ acting by translations, and $\Phi_\zeta = ds \wedge \varphi_\zeta + *\varphi_\zeta$ for a fibred G_2 -structure φ_ζ on P ; its rescaled version Φ_ζ^t (fibres scaled by t) is $ds \wedge \varphi_\zeta^t + *\varphi_\zeta^t$ [Maj26, Proposition 5.2, and the notation after Lemma 5.1]. Majewski's pre-glued Spin(7)-structure on $\mathbb{T} \times N$ is

$$\Phi_{\zeta;t}^{\text{pre}} := \Theta(\Phi_{t^2\zeta} \cup_t \Phi), \quad (2.8)$$

where $\cup_t$ interpolates using a cut-off function supported where the distance to S lies in $[2\epsilon, 3\epsilon]$, $\epsilon \sim t^\lambda$ for a fixed $\lambda \in [0, 1)$, and Θ is the projection onto Cayley forms; see [Maj26, Definition 6.1, (24)], [Maj26, §3.2.3] for the cut-off functions χ_i and [Maj26, §4.3] for the choice $\epsilon \sim t^\lambda$. In particular, since $\Phi_{t^2\zeta}$ is already a Cayley form,

$$\Phi_{\zeta;t}^{\text{pre}} = \Phi_{t^2\zeta} \quad \text{on } \{\text{dist}(\cdot, S) \leq 2\epsilon\} \supset \mathbb{T} \times P^R \quad \text{for } t \leq t(R), \quad (2.9)$$

because $\text{dist}(\cdot, S) = t\check{r}$ on P [JK21, §6.1] and $tR \leq 2\epsilon \sim 2t^\lambda$ for small t . We write φ_t^{pre} for the G_2 -structure on N with $\Phi_{\zeta;t}^{\text{pre}} = ds \wedge \varphi_t^{\text{pre}} + *\varphi_t^{\text{pre}}$.

The analytic core of [Maj26] is the family of weighted Hölder norms $\|\cdot\|_{\mathfrak{D}_{\beta;t}^{k+1,\alpha}}$ and $\|\cdot\|_{\mathfrak{E}_{\beta-1;t}^{k,\alpha}}$ on forms on the resolution [Maj26, Definition 4.7], [Maj25, Definition 9.1], which depend on a weight $\beta < 0$ not in a discrete set of critical rates, on $\alpha \in (0, 1)$, on λ , and on a further parameter κ [Maj26, Definition 4.5]. They are built from the weighted norms of [Maj25, Definition 8.1] on the ACF model (N_ζ, g_ζ^t) and the weighted norms of [Maj25, §7.0.2] on the orbifold; we recall the details needed here in the proof of Lemma 2.16. The main theorem is:

Theorem 2.10 ([Maj26, Theorem 6.1, Corollary 6.2]). *Suppose that the resolution $\rho_{\zeta;t}: (\mathbb{T} \times N, \Phi_{\zeta;t}^{\text{pre}}) \dashrightarrow (\mathbb{T} \times M', \Phi)$ is isentropic in the sense of [Maj26, Definition 4.10] (Assumption 8 of [Maj26]). Then there is $\vartheta > 0$, subject to the constraints listed before [Maj26, Proposition 6.1] and in the proof of [Maj26, Theorem 6.1], such that for all sufficiently small $t > 0$ there exists a torsion-free Spin(7)-structure $\Phi_{\zeta;t} = \Theta(\Phi_{\zeta;t}^{\text{pre}} + \eta_{\zeta;t})$ on $\mathbb{T} \times N$ with*

$$\|\Phi_{\zeta;t} - \Phi_{\zeta;t}^{\text{pre}}\|_{\mathfrak{D}_{\beta;t}^{1,\alpha}} \lesssim t^\vartheta; \quad (2.11)$$

here $\eta_{\zeta;t}$ is the unique solution of the fixed point problem [Maj26, (6)] in a suitable ball. If the pre-gluing data are $\mathbb{T}$ -invariant, then $\Phi_{\zeta;t} = ds \wedge \varphi_t + *\varphi_t$ for a torsion-free G_2 -structure φ_t on N .

For Joyce–Karigiannis data the isentropicity hypothesis is always satisfied. Indeed, by [Maj26, Corollary 4.1] (see also [Maj26, Remark 10]) it is equivalent to the degeneration at the second page of the Leray–Serre spectral sequence of $N_\zeta = \mathbb{T} \times P \rightarrow S = \mathbb{T} \times L$ [Maj26, Theorem 4.1, Proposition 4.3], whose second page is $H^\bullet(S) \oplus H^{\bullet-2}(S; Z)$: the only L^2 -harmonic forms on the Eguchi–Hanson fibres are the multiples of an anti-self-dual 2-form [Maj26, Lemma 4.2, Proposition 5.1], [Pla25, Theorem 3.26], and the corresponding local system on S is $\mathfrak{H}_\Gamma^2 \cong Z \times_{\mathbb{Z}_2} \mathbb{R}$. On the other hand P retracts onto Q , and $H^\bullet(Q) \cong H^\bullet(L) \oplus H^{\bullet-2}(L; Z)$ by [JK21, (6.19), §6.6]; so the total dimension of the second page equals $b_\bullet(N_\zeta)$ and the spectral sequence degenerates.

We now extract from Theorem 2.10 the statement that we shall use.

Theorem 2.12. *Let $(M', \varphi, g, L, \lambda)$ be Joyce–Karigiannis data and fix $\alpha \in (0, 1)$. There are $\theta > 0$, $t_0 > 0$, and for every $t \in (0, t_0)$ a torsion-free G_2 -structure φ_t on N with the following properties.*

- (i) For every compact $V \subset M' \setminus L = N \setminus Q$ there is $C_V > 0$ with $\|\varphi_t - \varphi\|_{C^{1,\alpha}(V,g)} \leq C_V t^\theta$.
(ii) For every $R \geq 1$ there is $C_R > 0$ with

$$\|t^{-3}(\varphi_t - \varphi_t^P)\|_{C^{1,\alpha}(P^R, \tilde{g}_t)} \leq C_R t^\theta. \quad (2.13)$$

- (iii) If K is a finite group acting on M' by automorphisms of (φ, g) preserving L and $\pm\lambda$, then K acts on N preserving Q and P^R , and φ_t can be chosen K -invariant.

Remark 2.14. Only $\theta > 0$ matters for [Theorem 1.1](#). Unwinding the constraints on ϑ in [\[Maj26, §6.1–§6.2\]](#) (with $m = 4$ the codimension of the stratum), one finds that (2.13) holds with θ slightly smaller than $\vartheta + \beta$, and that $\vartheta + \beta$ is bounded above by $\lambda + \alpha/2$ and by $\alpha/2 + (1 - \lambda)m - \lambda$ (from the third and fourth constraint in the list before [\[Maj26, Proposition 6.1\]](#), with the choice $\kappa = m/2 + \beta - \alpha/2$ of [\[Maj26, Remark 16\]](#)). Choosing $\lambda = 2/3$, $\beta \in (-2, 0)$ non-critical and α small, every $\theta \in (0, 2/3)$ is admissible; the choice $\lambda \approx 0.65$, $\vartheta \approx 2.6$ of [\[Maj26, Remark 16\]](#) corresponds to $\beta \approx -2$ and $\theta \approx 0.6$. In every case $\theta < 1$; see [Remark 1.4](#) for the significance of this. The appearance of β is explained in the proof of [Lemma 2.16](#): near Q the weight function of the norm $\mathfrak{D}_{\beta;t}^{1,\alpha}$ is of size t , so that (2.11) controls the difference $\Phi_{\zeta;t} - \Phi_{\zeta;t}^{\text{pre}}$ near Q only up to $t^{\vartheta+\beta}$.

Remark 2.15. The torsion-free G_2 -structures φ_t of [Theorem 2.12](#) need not coincide with the structures $\tilde{\varphi}_t^N$ of [\[JK21, Theorem 6.4\]](#), which are obtained from a different pre-glued structure φ_t^N [\[JK21, \(6.20\)\]](#) via [\[Joy00, Theorem 11.6.1\]](#); the two families lie in the same connected component of the moduli space of torsion-free G_2 -structures on N for small t , but we shall not need this.

The proof of [Theorem 2.12](#) occupies the rest of this section. The first lemma makes the weighted estimate (2.11) explicit near Q .

Lemma 2.16. *Let $R \geq 1$ and $\alpha' \in (0, \alpha]$. There is $C = C(R, \alpha') > 0$ such that for all t small and every $\mathbb{T}$ -invariant 4-form $\Psi = ds \wedge \psi + \psi'$ on $\mathbb{T} \times N$,*

$$\|t^{-3}\psi\|_{C^{1,\alpha'}(P^R, \tilde{g}_t)} \leq C t^{\beta-\alpha'(\alpha-\beta)} \|\Psi\|_{\mathfrak{D}_{\beta;t}^{1,\alpha}}. \quad (2.17)$$

Proof. We recall the definition of the norms. By [\[Maj26, Definition 4.7\]](#) (= [\[Maj25, Definition 9.1\]](#)),

$$\|\Psi\|_{\mathfrak{D}_{\beta;t}^{1,\alpha}} = \|\delta_t^*((1 - \chi_4)\Psi)\|_{\mathfrak{D}_{\text{ACF};\beta;t}^{1,\alpha}} + \|\chi_2\Psi\|_{C_{\text{CFS};\beta;\epsilon}^{1,\alpha}},$$

where χ_4 vanishes on $\{\text{dist}(\cdot, S) \leq 3\epsilon\} \supset \mathbb{T} \times P^R$ (for t small) and δ_t identifies the resolution region with the fixed manifold $N_\zeta = \mathbb{T} \times P$ carrying the metrics $g_\zeta^t = ds^2 + g_t^P$ [\[Maj26, §3.2.3, Remark 6\]](#). By [\[Maj26, Definition 4.5, \(11\)\]](#),

$$\|\Xi\|_{\mathfrak{D}_{\text{ACF};\beta;t}^{1,\alpha}} = \|\pi_{\mathcal{COI};\beta}\Xi\|_{C_{\text{ACF};\beta;t}^{1,\alpha}} + t^{-\kappa}\|\varpi_{\mathcal{K};\beta}\Xi\|_{C_t^{1,\alpha}},$$

where $\varpi_{\mathcal{K};\beta}$ is the projection onto the fibrewise L^2 -harmonic part and $\pi_{\mathcal{COI};\beta}$ its complement, $\|\cdot\|_{C_t^{1,\alpha}}$ is the unweighted Hölder norm of g_ζ^t , and $\kappa > \beta$ for the choice $\kappa = m/2 + \beta - \alpha/2$ of [\[Maj26, Remark 16\]](#), which we adopt. Finally, by [\[Maj25, Definition 8.1, Remark 39\]](#) the norm $\|\cdot\|_{C_{\text{ACF};\beta;t}^{1,\alpha}}$ is the weighted Hölder norm on (N_ζ, g_ζ^t) with weight function $w_{\text{ACF};t} = t \cdot w_{\text{ACF}}$, where w_{ACF} is a positive function on N_ζ asymptotic to the radius function; the weight of a term with i horizontal and $j - i$ vertical derivatives is $w_{\text{ACF};t}^{-(\beta-j+i)}$, i.e. each vertical derivative lowers the weight exponent by one, and the Hölder seminorms are taken over pairs of points at g_ζ^t -distance at most $w_{\text{ACF};t}^{-(\beta-j+i-\alpha)}$. On $\mathbb{T} \times P^R$ we have $c_R \leq w_{\text{ACF}} \leq C_R$, hence $c_R t \leq w_{\text{ACF};t} \leq C_R t$.

Now let Ξ be a 4-form on $\mathbb{T} \times P^R$ and compare the norms of g_ζ^t with those of the rescaled metric $t^{-2}g_\zeta^t = t^{-2}ds^2 + \tilde{g}_t$. For the $ds \wedge \psi$ -component the pointwise norms with respect to product metrics reduce to those of ψ , and $\nabla(ds \wedge \psi) = ds \wedge \nabla\psi$; so it suffices to consider ψ

on P^R . For a 3-form, $|t^{-3}\psi|_{\tilde{g}_t} = |\psi|_{g_t^P}$. A $\tilde{g}_t$ -unit horizontal vector is t times a g_L -unit vector, and a $\tilde{g}_t$ -unit vertical vector v has g_t^P -length t ; consequently

$$|\nabla_{\text{hor}}(t^{-3}\psi)|_{\tilde{g}_t} = t |\nabla_{\text{hor}}\psi|_{g_t^P}, \quad |\nabla_v(t^{-3}\psi)|_{\tilde{g}_t} = t |\nabla_{v/t}\psi|_{g_t^P},$$

where on the right hand side the derivatives are with respect to g_L -unit horizontal, respectively g_t^P -unit vertical, vectors. Since $\Xi := \pi_{\mathcal{CO}\mathcal{I};\beta}\Psi$ satisfies $|\Xi| \leq w_{\text{ACF};t}^\beta \|\Xi\|$, $|\nabla_{\text{hor}}\Xi| \leq w_{\text{ACF};t}^\beta \|\Xi\|$ and $|\nabla_{\text{vert}}\Xi| \leq w_{\text{ACF};t}^{\beta-1} \|\Xi\|$ (norms with respect to g_ζ^t , and $\|\Xi\| := \|\Xi\|_{C_{\text{ACF};\beta;t}^{1,\alpha}}$), we obtain on P^R

$$|t^{-3}\Xi|_{\tilde{g}_t} + |\nabla(t^{-3}\Xi)|_{\tilde{g}_t} \leq C_R t^\beta \|\Xi\|.$$

The Hölder seminorms of $t^{-3}\Xi$ and $\nabla(t^{-3}\Xi)$ are controlled in the same way (with the same power t^β) for pairs of points at $\tilde{g}_t$ -distance at most $\delta := ct^{\alpha-\beta}$, which is the rescaled version of the constraint $\text{dist}_{g_\zeta^t}(x, y) \leq w_{\text{ACF};t}^{-(\beta-1-\alpha)}$ near Q ; for pairs at distance between δ and 1 we use the C^0 -bound, which costs a factor $\delta^{-\alpha'} = ct^{-\alpha'(\alpha-\beta)}$. This gives (2.17) for $\pi_{\mathcal{CO}\mathcal{I};\beta}\Psi$. For the fibrewise harmonic part $\varpi_{\mathcal{K};\beta}\Psi$ the same conversion applied to the unweighted norm $\|\cdot\|_{C_t^{1,\alpha}}$ gives a bound by $C_R t^\beta \|\Psi\|_{\mathfrak{D}_{\beta;t}^{1,\alpha}} \leq C_R t^\beta \|\Psi\|_{\mathfrak{D}_{\beta;t}^{1,\alpha}}$. $\square$

The second lemma compares Majewski's local model with the fibred model (2.4).

Lemma 2.18. *For every $R \geq 1$ there is $C_R > 0$ such that for all small $t > 0$*

$$\|t^{-3}(\varphi_t^{\text{pre}} - \varphi_t^P)\|_{C^{1,\alpha}(P^R, \tilde{g}_t)} \leq C_R t. \quad (2.19)$$

Proof. By (2.9), $\varphi_t^{\text{pre}} = \varphi_\zeta^t$ on P^R . Unwinding [Maj26, §5.3.2] (the universal Cayley form $\Phi_{\mathfrak{N}_\Gamma} = \Phi^{4,0} + \Phi^{2,2} + \Phi^{0,4}$ with $\Phi^{4,0} = \text{vol}_S$, $\Phi^{2,2} = -\langle [\theta_S \wedge \theta_S]_+ \wedge [\underline{\omega}_{\mathcal{M}_\Gamma}] \rangle$, $\Phi^{0,4} = \frac{1}{6} \langle [\underline{\omega}] \wedge [\underline{\omega}] \rangle$, and the definition $(N_\zeta, \Phi_\zeta) = \zeta^*(\mathfrak{N}_\Gamma, \Phi_{\mathfrak{N}_\Gamma})$ and the identification $\Lambda_+^2 T^*(\mathbb{T} \times L) \cong T^*L$, $\text{ds} \wedge e_a + *_L e_a \leftrightarrow e_a$, one sees that φ_ζ^t is of the form (2.4): it is built from the same base metric g_L , the same coframe (e_1, e_2, e_3) , and the same fibrewise Eguchi–Hanson triples (with parameter $A(x)$ and complex structure I , by our normalisation of ζ), but with respect to the splitting $TP = \check{V} \oplus \check{H}_\zeta$ induced by the Ehresmann connection H_ζ of [Maj26, proof of Theorem 5.1], which comes from the Levi-Civita connection of g on the frame bundle of ν together with the connection of the universal family. The splitting used in (2.4) comes from $\check{\nabla}$. Both are induced by connections on the principal bundle underlying ν (with structure group $\text{U}(2)$, respectively the normaliser of Γ in $\text{SO}(4)$, cf. [JK21, (4.4)] and [Maj26, §5.3.2]), hence differ by a 1-form B on L with values in the corresponding Lie algebra bundle, which is bounded in C^1 on L together with its derivatives. Changing the splitting by B changes a form of type $(2, 0)$ such as $\check{\omega}^I$ by a form of type $(1, 1)$ of the form $B \wedge \iota_{\xi_B} \check{\omega}^I$, where ξ_B is the vertical vector field generated by B , plus a term of type $(0, 2)$ quadratic in B . In the rescaled picture (2.5), i.e. in coordinates $\tilde{y} = y/t$ on L , the connection 1-form becomes $tB(t\tilde{y})$, whose C^1 -norm on unit balls is $O(t)$. Since $\tilde{\varphi}_t^P$ involves $\check{\omega}^I, \check{\omega}^J, \check{\omega}^K$ linearly with coefficients of size one in $\tilde{g}_t$, the difference of the two rescaled G_2 -structures on P^R is $O(t)$ in $C^{1,\alpha}(\tilde{g}_t)$, uniformly on P^R because ξ_B is bounded there. $\square$

Lemma 2.20. *Let K be as in Theorem 2.12 (iii). Then K acts on N by diffeomorphisms preserving Q , P^R , φ_t^P , g_t^P , φ_t^{pre} , and the torsion-free G_2 -structure φ_t of Theorem 2.10 can be chosen K -invariant.*

Proof. All ingredients of the construction of N , Q , (P, φ_t^P, g_t^P) and φ_t^{pre} are natural: ν , h_ν , ∇^ν , I (up to sign, which does not matter [JK21, §6.6]), the normal exponential map Υ , the distance function to L , the cut-off functions of [Maj26, §3.2.3] (which are functions of the distance to S), and the universal family of [Maj26, Proposition 5.1], which is equivariant under the normaliser of Γ . Hence K acts on $\mathbb{T} \times N$ preserving all of these. The only non-canonical choice in the proof of [Maj26, Theorem 6.1] is the right inverse $R_{\Phi^{\text{pre}}}$ of the Hodge–de Rham operator $D_{\Phi^{\text{pre}}}$ from [Maj26, Corollary 6.1], which enters the fixed point map $\mathcal{C}_t$ of [Maj26,

proof of Theorem 6.1]. Since $D_{\Phi^{\text{pre}}}$ and the norms $\mathfrak{D}_{\beta;t}^{1,\alpha}$, $\mathfrak{C}_{\beta-1;t}^{0,\alpha}$ are K -invariant, the average $\frac{1}{|K|} \sum_{k \in K} k^* R_{\Phi^{\text{pre}}}(k^{-1})^*$ is again a bounded right inverse with the same bounds, and we use it instead. Then $\mathcal{C}_t$ commutes with K , so its unique fixed point $\eta_{\zeta;t}$ in the relevant ball is K -invariant, and so is $\Phi_{\zeta;t} = \Theta(\Phi^{\text{pre}} + \eta_{\zeta;t})$. The same argument with $\mathbb{T}$ in place of K shows $\mathbb{T}$ -invariance, as used in [Maj26, §6.2.1]. $\square$

Proof of Theorem 2.12. As explained above, the isentropicity hypothesis of Theorem 2.10 holds, so we obtain $\Phi_{\zeta;t} = ds \wedge \varphi_t + * \varphi_t$ with (2.11), and φ_t is K -invariant by Lemma 2.20; this proves (iii). By (2.9), Lemma 2.16 applied to $\Psi = \Phi_{\zeta;t} - \Phi_{\zeta;t}^{\text{pre}}$ (and $\alpha' = \alpha$) and Lemma 2.18,

$$\|t^{-3}(\varphi_t - \varphi_t^P)\|_{C^{1,\alpha}(P_R, \tilde{g}_t)} \leq C_R(t^{\vartheta+\beta-\alpha(\alpha-\beta)} + t) \leq C_R t^\theta$$

with $\theta := \min\{\vartheta + \beta - \alpha(\alpha - \beta), 1\}$, which is positive for the parameters of Remark 2.14. This proves (ii). Finally, on a compact subset $V \subset M' \setminus L$ the pre-glued structure agrees with Φ for t small, and the weight function of the norm $C_{\text{CFS};\beta;\epsilon}^{1,\alpha}$ in [Maj26, Definition 4.7] is bounded above and below on V [Maj25, §7.0.2]; hence (2.11) gives (i). $\square$

3. RIGID EQUIVARIANT ASSOCIATIVE SUBMANIFOLDS

3.1. An equivariant perturbation result. We use the deformation theory of associative submanifolds in the form of [DPW23, §3]. Recall from [DPW23, Definition 3.7, Proposition 3.9]: given a G_2 -structure φ on a 7-manifold Y , a closed φ -associative submanifold $P_0 \subset Y$ and a tubular neighbourhood $\mathbf{j}: U \rightarrow Y$, $U \subset NP_0$, there is a chart $\mathbf{Q} = \mathbf{Q}_{\mathbf{j}}: \Gamma(U) \rightarrow \mathcal{S}$, $v \mapsto \mathbf{j}(\Gamma_v)$, of the space $\mathcal{S}$ of submanifolds near P_0 , and a smooth map $f: \Gamma(U) \rightarrow \Gamma(NP_0)$ such that $\mathbf{Q}(v)$ is φ -associative if and only if $f(v) = 0$; moreover

$$f(v) = D_{P_0}^\varphi v + \mathcal{N}(v), \quad \|\mathcal{N}(v) - \mathcal{N}(w)\|_{C^{0,\alpha}} \leq c(\|v\|_{C^{1,\alpha}} + \|w\|_{C^{1,\alpha}})\|v - w\|_{C^{1,\alpha}}, \quad (3.1)$$

where c depends on $\mathbf{j}$ and the geometry of (Y, g_φ) on $\mathbf{j}(U)$ [DPW23, Proposition 3.9]. Explicitly, $\langle f(v), w \rangle_{L^2} = \int_{\Gamma_v} i_w \mathbf{j}^* \psi$ with $\psi = *\varphi$ [DPW23, proof of Proposition 3.9]; in particular f depends smoothly on the 1-jet of ψ along $\mathbf{j}(U)$. The following is the rigid, equivariant counterpart of [DPW23, Proposition 3.18]; cf. [DPW23, 4.2].

Proposition 3.2. *Let $\alpha \in (0, 1)$ and $\theta, c_1, c_2, c_3, R > 0$. There are $T = T(\alpha, \theta, c_1, c_2, c_3, R) > 0$ and $c_v = c_v(c_1, c_2, c_3) > 0$ with the following significance. Let φ_0, φ be G_2 -structures on a 7-manifold Y , let K be a finite group acting on Y preserving φ_0 and φ , let $P_0 \subset Y$ be a closed K -invariant φ_0 -associative submanifold, and let $\mathbf{j}: U \rightarrow Y$ be a K -equivariant tubular neighbourhood of P_0 . Denote by $f_0, f: \Gamma(U) \rightarrow \Gamma(NP_0)$ the maps (3.1) for φ_0 and φ , by $D_0 := D_{P_0}^{\varphi_0}$, and by $\Gamma(NP_0)^K$ the K -invariant sections; Hölder norms are taken with respect to g_{φ_0} . Let $t \in (0, T)$ and suppose that:*

- (1) $B_R(0) \subset U$.
- (2) $\|\mathbf{j}^*(\varphi - \varphi_0)\|_{C^{1,\alpha}(U)} \leq c_1 t^\theta$.
- (3) D_0 maps $C^{1,\alpha}\Gamma(NP_0)^K$ bijectively onto $C^{0,\alpha}\Gamma(NP_0)^K$, and $\|v\|_{C^{1,\alpha}} \leq c_2 \|D_0 v\|_{C^{0,\alpha}}$ for $v \in C^{1,\alpha}\Gamma(NP_0)^K$.
- (4) The estimate (3.1) for φ_0 holds with constant c_3 on $B_R(0)$, and for $v, w \in B_R(0) \subset C^{1,\alpha}\Gamma(NP_0)$

$$\|(f - f_0)(v) - (f - f_0)(w)\|_{C^{0,\alpha}} \leq c_3 \|\mathbf{j}^*(\varphi - \varphi_0)\|_{C^{1,\alpha}(U)} \|v - w\|_{C^{1,\alpha}},$$

$$\|(f - f_0)(0)\|_{C^{0,\alpha}} \leq c_3 \|\mathbf{j}^*(\varphi - \varphi_0)\|_{C^{1,\alpha}(U)}.$$

Then there is a unique $v \in \Gamma(U)^K$ with $\|v\|_{C^{1,\alpha}} \leq c_v t^\theta$ such that $\mathbf{Q}(v)$ is φ -associative; moreover v is the unique K -invariant solution of $f(v) = 0$ in the ball of radius $r_0 := 1/(8c_2 c_3)$.

Proof. Since $\mathbf{j}$, φ_0 and φ are K -invariant, the maps $f_0, f, \mathcal{N}$ and D_0 are K -equivariant, so they map K -invariant sections to K -invariant sections. For $v \in C^{1,\alpha}\Gamma(NP_0)^K$, by (3) the equation $f(v) = 0$ is equivalent to $v = -E(v)$ with

$$E(v) := D_0^{-1}(\mathcal{N}(v) + (f - f_0)(v)) \in C^{1,\alpha}\Gamma(NP_0)^K.$$

By (2)–(4), for v, w in the ball of radius $r \leq r_0$ in $C^{1,\alpha}\Gamma(NP_0)^K$,

$$\|E(0)\|_{C^{1,\alpha}} \leq c_2 c_3 c_1 t^\theta, \quad \|E(v) - E(w)\|_{C^{1,\alpha}} \leq c_2 c_3 (2r + c_1 t^\theta) \|v - w\|_{C^{1,\alpha}} \leq \frac{1}{2} \|v - w\|_{C^{1,\alpha}}$$

provided $c_1 c_2 c_3 t^\theta \leq 1/4$. Hence $-E$ is a contraction of the ball of radius $r := 2c_2 c_3 c_1 t^\theta$ into itself as soon as $r \leq r_0$ and $r \leq R$, which holds for $t < T$; its unique fixed point v has $\|v\|_{C^{1,\alpha}} \leq c_v t^\theta$ with $c_v := 2c_2 c_3 c_1$, and it is also the unique fixed point in the ball of radius r_0 since $-E$ is a contraction there. By [DPW23, Proposition 3.9] and elliptic regularity v is smooth. $\square$

3.2. The model associative submanifolds. We now return to the situation of Theorem 1.1, so let $L_0, \ell, (W, \omega, \Omega), \Sigma, U, K, K_W, c$ be as in Theorem 1.1 (a), (b). Using (1.2) we identify U with $(\mathbb{R}/\ell\mathbb{Z}) \times U_W$ and L_0 with $(\mathbb{R}/\ell\mathbb{Z}) \times \Sigma$.

Lemma 3.3. *In the situation of Theorem 1.1:*

- (i) $g|_{L_0} = dx^2 + g_\Sigma$, where g_Σ is the metric induced by (ω, Ω) on Σ ; $A = |\lambda| = 1$ on L_0 , and $e_1 = \pm dx$.
- (ii) $\nu|_{L_0}, h_\nu$ and ∇^ν are the pull-backs of the normal bundle $N_{\Sigma/W}$ of Σ in W with its induced metric and connection, and $I = \pm J_W$, where J_W is the complex structure of W ; in particular $\nabla^\nu I = 0$, and we may and do take $\check{\nabla} = \nabla^\nu$ in Section 2.2.
- (iii) $\sigma^{-1}(L_0) = (\mathbb{R}/\ell\mathbb{Z}) \times P_W$ and $Q|_{L_0} = (\mathbb{R}/\ell\mathbb{Z}) \times Q_W$, where $\sigma_W: P_W \rightarrow \Sigma$ is the fibrewise blow-up of $N_{\Sigma/W}/\{\pm 1\}$ along the zero section with respect to J_W and $Q_W = \mathbb{P}(N_{\Sigma/W}) \rightarrow \Sigma$ is the exceptional $\mathbb{CP}^1$ -bundle; φ_t^P and g_t^P are invariant under the translations of $\mathbb{R}/\ell\mathbb{Z}$.
- (iv) $Q_c = (\mathbb{R}/\ell\mathbb{Z}) \times Q_{W,c} \cong S^1 \times S^2$ is φ_t^P -associative (for a suitable orientation), and $(Q_c, \tilde{g}_t|_{Q_c})$ is isometric to the product of $\mathbb{R}/t^{-1}\ell\mathbb{Z}$ with a round 2-sphere of area π .
- (v) K acts on $\sigma^{-1}(L_0) = (\mathbb{R}/\ell\mathbb{Z}) \times P_W$ by $\text{id} \times K_W$, preserves Q_c , and acts on $T_c\Sigma$ without non-zero fixed vectors.

Proof. (i) and (ii) follow from (1.2): the metric of $dx \wedge \omega + \text{Re } \Omega$ is $dx^2 + g_W$ [JK21, §2.6], and for $v, w \perp \partial_x$ we have $\varphi(\partial_x, v, w) = \omega(v, w) = g_W(J_W v, w)$, so that $\partial_x \times v = J_W v$. For (iii) note that P, Q and the data $\check{\omega}^I, \check{\omega}^J, \check{\omega}^K, \check{g}_{2,0}$ are constructed fibrewise from $(\nu, h_\nu, I, \check{\nabla})$ and (e_1, e_2, e_3) , which are pulled back from Σ ; the coframe may be chosen as $e_1 = dx$ and (e_2, e_3) a local oriented orthonormal coframe of (Σ, g_Σ) . For (iv): the tangent space of Q_c at a point of the fibre $Q_x, x = (s, c)$, is $\langle \tilde{e}_1 \rangle \oplus TQ_x$, where $\tilde{e}_1$ is the $\check{H}$ -horizontal lift of ∂_x ; indeed Q is preserved by the parallel transport of $\check{\nabla}$ since $\check{\nabla} I = 0$, cf. [JK21, (4.4)]. For $v, w \in TQ_x$, (2.4) gives $\varphi_t^P(\tilde{e}_1, v, w) = -t^2 \check{\omega}^I(v, w)$, because $\sigma^*(e_2)$ and $\sigma^*(e_3)$ vanish on $\tilde{e}_1$ and $\check{\omega}^J, \check{\omega}^K$ vanish on Q_x (Section 2.2); moreover $\check{\omega}^I|_{Q_x}$ is the area form of Q_x for the metric $\check{g}_{2,0}$ up to sign, as Q_x is I -holomorphic. Since $\tilde{e}_1$ is g_t^P -orthogonal to TQ_x and of length one, $|\varphi_t^P|_{Q_c}| = \text{vol}_{Q_c}$, i.e. Q_c is calibrated by φ_t^P with a suitable orientation. The isometry statement follows from $\tilde{g}_t = t^{-2} \sigma^*(g_L) + \check{g}_{2,0}$, $A = 1$ and [JK21, Remark 2.10], which identifies the bolt of the Eguchi–Hanson space (X, h_1) with the round sphere of radius $\frac{1}{2}$. (v) follows from the naturality of the construction (Lemma 2.20) and (b). $\square$

3.3. The linearised operator. Let $(X, h_1, \omega_1^I, \omega^J, \omega^K)$ be the Eguchi–Hanson space with parameter $a = 1$ and bolt $Y \subset X$, and consider the flat model

$$Y_0 := (\mathbb{R}/t^{-1}\ell\mathbb{Z}) \times \{0\} \times Y \subset (\mathbb{R}/t^{-1}\ell\mathbb{Z}) \times T_c\Sigma \times X \quad (3.4)$$

with the G_2 -structure (1.6), where $(\mathbb{R}/t^{-1}\ell\mathbb{Z}) \times T_c\Sigma$ carries the flat metric $dx^2 + g_\Sigma|_c$ and we use the coordinates $x_1 = x$ and (x_2, x_3) on $T_c\Sigma$ dual to $(e_2, e_3)|_c$. This is the situation of [DPW23, 3.2] (with $\mu = \text{id}$, $\Sigma = Y \cong S^2$, $L = t^{-1}\ell$), and Y_0 is associative. Its normal bundle is $NY_0 = T_c\Sigma \oplus NY$, and by [DPW23, 3.12, 3.21] the Fueter operator is

$$D_{\text{prod}} = (-i \oplus I) \cdot (\partial_x + A), \quad A = \begin{pmatrix} 0 & i\bar{\partial}^* \kappa^* \\ -\kappa \bar{\partial} i & 0 \end{pmatrix}, \quad (3.5)$$

where A is a formally self-adjoint elliptic operator on $V := T_c\Sigma \oplus NY \rightarrow Y$ with

$$\ker A = T_c\Sigma \quad (\text{the constant sections of the trivial summand}), \quad (3.6)$$

because $H^{0,1}(S^2; \mathbb{C}) = 0$ [DPW23, 3.12]. In particular $\ker D_{\text{prod}} = T_c\Sigma$ consists of the constant sections, corresponding to translations of Y_0 in $T_c\Sigma$.

Lemma 3.7. *There is $C > 0$, independent of t , and there are identifications of a tubular neighbourhood of Q_c in $(P^R, \tilde{g}_t)$ with a tubular neighbourhood of Y_0 in the flat model (3.4), and of NQ_c with NY_0 , under which*

$$\|D_{Q_c}^{\tilde{\varphi}_t^P} - D_{\text{prod}}\|_{C^{1,\alpha} \rightarrow C^{0,\alpha}} \leq Ct.$$

Proof. Choose Riemannian normal coordinates $y = (y_2, y_3)$ for (Σ, g_Σ) centred at c , and trivialise $\nu|_{L_0} = \text{pr}^*N_{\Sigma/W}$ over the corresponding disc by radial parallel transport with respect to $\tilde{\nabla} = \nabla^\nu$ (Lemma 3.3 (ii)), so that the connection 1-form B of ∇^ν satisfies $B(y) = O(|y|)$ and $\nabla B = O(1)$. By Lemma 3.3 this trivialises $\sigma^{-1}(\mathbb{R}/\ell\mathbb{Z} \times D) \cong (\mathbb{R}/\ell\mathbb{Z}) \times D \times X$ in a way which is independent of x , and under this identification (φ_t^P, g_t^P) is given by (2.4) with $e_1 = dx$, (e_2, e_3) the coframe of g_Σ in the coordinates y , the fibrewise data equal to those of $(X, h_1, \omega_1^I, \omega^J, \omega^K)$ (since $A \equiv 1$), and the horizontal distribution given by the graph of $-B$. Pass to the rescaled coordinates $\tilde{x} = x/t$, $\tilde{y} = y/t$ of (2.5). The coefficients of g_Σ become $\delta_{ij} + O(t^2|\tilde{y}|^2)$ with $\tilde{y}$ -derivatives $O(t^2|\tilde{y}|)$, and the connection 1-form becomes $tB(t\tilde{y})d\tilde{y} = O(t^2|\tilde{y}|)$ with derivatives $O(t^2)$. Hence on $\{|\tilde{y}| \leq 1, \tilde{r} \leq R\}$ the rescaled G_2 -structure $\tilde{\varphi}_t^P$ and its metric $\tilde{g}_t$ agree with the flat model (1.6) on $(\mathbb{R}/t^{-1}\ell\mathbb{Z}) \times T_c\Sigma \times X$ up to $O(t^2)$ in C^1 , uniformly in t (note that all data are $\tilde{x}$ -independent). Since the Fueter operator $D = -\gamma\nabla + \tau^\perp$ of [DPW23, Definition 3.8] is determined by the 1-jet of the G_2 -structure along the submanifold (the cross product γ by φ , the normal connection ∇ by the 1-jet of the metric, and the torsion τ by the 1-jet of ψ), the claim follows, with the identifications of tubular neighbourhoods and normal bundles given by the coordinates $(\tilde{x}, \tilde{y}, p)$. $\square$

The following is the variant of [DPW23, Proposition 3.22] in which the loss $(L+1)$ caused by the length L of the circle is absent.

Lemma 3.8. *In the situation of [DPW23, Situation 3.20], let X be a compact oriented Riemannian manifold, $V \rightarrow X$ a Euclidean vector bundle, $A: \Gamma(V) \rightarrow \Gamma(V)$ a formally self-adjoint linear elliptic differential operator of first order, and $\pi: \Gamma(V) \rightarrow \ker A$ the L^2 -orthogonal projection. For $\alpha \in (0, 1)$ there is $c = c(A, \alpha) > 0$ such that for every $L > 0$ and every $s \in C^{1,\alpha}\Gamma((\mathbb{R}/L\mathbb{Z}) \times X, V)$ with*

$$\pi(i_x^*s) = 0 \quad \text{for all } x \in \mathbb{R}/L\mathbb{Z}, \quad (3.9)$$

where $i_x(p) := (x, p)$, one has

$$\|s\|_{C^{1,\alpha}} \leq c \|(\partial_x + A)s\|_{C^{0,\alpha}}.$$

Proof. This is contained in the proof of [DPW23, Proposition 3.22]: with $\hat{\pi}s(x, p) := (\pi(i_x^*s))(p)$, one has $\|s\|_{C^{1,\alpha}} \leq c_1(\|(\partial_x + A)s\|_{C^{0,\alpha}} + \|s\|_{L^\infty})$ by interior Schauder estimates and $\|(\mathbf{1} - \hat{\pi})s\|_{L^\infty} \leq c_2\|(\partial_x + A)(\mathbf{1} - \hat{\pi})s\|_{C^{0,\alpha}}$ by a contradiction argument, with c_1, c_2 independent of L (cf. [Wal13, proof of Proposition 8.5]); the hypothesis (3.9) means $\hat{\pi}s = 0$. For the reader's convenience we recall the contradiction argument. If the second estimate failed there would be $L_n > 0$ and s_n with $\hat{\pi}s_n = 0$, $\|s_n\|_{L^\infty} = 1$ and $\|(\partial_x + A)s_n\|_{C^{0,\alpha}} \rightarrow 0$; after translating in x we may assume $|s_n(0, p_n)| \geq \frac{1}{2}$ for some $p_n \in X$. By the Schauder estimates a subsequence converges in C^1 on compact subsets (of $\mathbb{R} \times X$ if $L_n \rightarrow \infty$, or of a circle otherwise) to a bounded solution s of $(\partial_x + A)s = 0$ with $\hat{\pi}s = 0$ and $s \neq 0$. Expanding $s(x, \cdot)$ in an L^2 -orthonormal eigenbasis (ϕ_μ) of A gives $s(x, \cdot) = \sum_\mu a_\mu(0)e^{-\mu x}\phi_\mu$, and boundedness forces $a_\mu(0) = 0$ for $\mu \neq 0$, so $s = \hat{\pi}s = 0$, a contradiction. $\square$

Lemma 3.10. *Let K act on $Y_0 = (\mathbb{R}/t^{-1}\ell\mathbb{Z}) \times Y$ and on $NY_0 = T_c\Sigma \oplus NY$ as in Lemma 3.3 (v), i.e. trivially on the circle and via K_W on $T_c\Sigma$ and on (Y, NY) . Then every K -invariant section*

s of NY_0 satisfies (3.9) for the operator A of (3.5), and D_{prod} maps $C^{1,\alpha}\Gamma(NY_0)^K$ bijectively onto $C^{0,\alpha}\Gamma(NY_0)^K$ with $\|s\|_{C^{1,\alpha}} \leq c\|D_{\text{prod}}s\|_{C^{0,\alpha}}$, where c is independent of t .

Proof. K acts on Y by isometries which are holomorphic for I and preserve the flat model (3.4) and Y_0 ; hence A and D_{prod} are K -equivariant, and so is the projection $\hat{\pi}$ onto the x -dependent constant sections of $T_c\Sigma$ ((3.6); the projection is the fibrewise average of the $T_c\Sigma$ -component). For K -invariant s and each x , $\pi(i_x^*s) \in (T_c\Sigma)^K = 0$ by Theorem 1.1 (b), which is (3.9). The estimate now follows from Lemma 3.8, since $-i \oplus I$ is an isometry. The operator D_{prod} is formally self-adjoint [DPW23, 3.10], so its restriction to K -invariant sections has $\ker = \text{coker}$; the estimate shows that the kernel is trivial and the range is closed, hence the restriction is bijective. $\square$

3.4. Proof of Theorem 1.1. Let $R \geq 2$ and let t be small. We apply Proposition 3.2 with $Y := P^R \subset N$ (or rather a tubular neighbourhood of Q_c in P^R), $\varphi_0 := \tilde{\varphi}_t^P = t^{-3}\varphi_t^P$, $\varphi := t^{-3}\varphi_t$, $P_0 := Q_c$, the group K , and $\mathfrak{j}$ the normal exponential map of $\tilde{g}_t$ restricted to a ball bundle of fixed radius R' in NQ_c ; $\mathfrak{j}$ is K -equivariant because K acts isometrically on $(P^R, \tilde{g}_t)$ and preserves Q_c (Lemma 2.20, Lemma 3.3 (v)). Note that $\mathbf{Q}(v)$ is φ -associative if and only if it is φ_t -associative. We check the hypotheses, with all norms taken with respect to $\tilde{g}_t$, whose geometry on P^R is bounded uniformly in t (Section 2.2), so that R' and hypothesis (1) can be arranged with R independent of t .

- (2) holds by Theorem 2.12 (ii), with $c_1 = C_R$ and the exponent θ of Theorem 2.12.
- (3): by Lemma 3.7 and Lemma 3.10, for K -invariant v we have $\|v\|_{C^{1,\alpha}} \leq c\|D_{\text{prod}}v\| \leq c\|D_0v\|_{C^{0,\alpha}} + cCt\|v\|_{C^{1,\alpha}}$, whence $\|v\|_{C^{1,\alpha}} \leq 2c\|D_0v\|_{C^{0,\alpha}}$ for $t \leq (2cC)^{-1}$; and D_0 is a small perturbation of the bijection D_{prod} on K -invariant sections, hence bijective. Thus (3) holds with $c_2 = 2c$.
- (4) holds by (the proof of) [DPW23, Proposition 3.9], applied in the bounded geometry of $(P^R, \tilde{g}_t)$; cf. [DPW23, proof of Proposition 4.1], where the same point is made using [DPW23, 2.18]. The constants only depend on bounds for the geometry of $\tilde{g}_t$ and on R' , hence are independent of t .

Proposition 3.2 yields, for $t < t_1$, a unique K -invariant v_t with $\|v_t\|_{C^{1,\alpha}} \leq c_v t^\theta$ such that $P_t := \mathbf{Q}(v_t)$ is φ_t -associative, and the uniqueness statement with $r_0 = 1/(8c_2c_3)$. Clearly P_t is K -invariant and diffeomorphic to $Q_c \cong S^1 \times S^2$ (Lemma 3.3 (iv)), and $[P_t] = [Q_c]$ since P_t is isotopic to Q_c . Since $\|v_t\|_{C^1} \leq c_v t^\theta$, the volume of P_t with respect to $\tilde{g}_t$ is $\text{vol}_{\tilde{g}_t}(Q_c)(1 + O(t^\theta)) = \pi t^{-1}\ell(1 + O(t^\theta))$ by Lemma 3.3 (iv), and $g_{\varphi_t} = t^2g_\varphi = t^2\tilde{g}_t(1 + O(t^\theta))$ on P^R by Theorem 2.12 (ii); hence $\text{vol}_{g_{\varphi_t}}(P_t) = \pi\ell t^2(1 + O(t^\theta))$. Finally, $P_t \subset \mathfrak{j}(B_{c_v t^\theta}(0))$ lies within $\tilde{g}_t$ -distance $c_v t^\theta$ of Q_c , i.e. within g_t^P -distance $c_v t^{1+\theta}$; as Q_c collapses onto the circle $(\mathbb{R}/\ell\mathbb{Z}) \times \{c\}$ in (M', g) as $t \rightarrow 0$, so does P_t . This completes the proof of Theorem 1.1. $\square$

4. EXAMPLE: A RESOLUTION OF $(T^3 \times K3)/\mathbb{Z}_2^2$

4.1. The orbifold of [JK21, Example 7.2]. Let $C = \{[z_0, z_1, z_2] \in \mathbb{CP}^2 : F(z) = 0\}$ with $F(z) := z_0^6 + z_1^6 + z_2^6$ be the Fermat sextic, a smooth curve of genus 10, and let $\varpi: X \rightarrow \mathbb{CP}^2$ be the double cover branched along C ; thus $X = \{w^2 = F(z)\}$ in the total space of $\mathcal{O}_{\mathbb{CP}^2}(3)$, and X is a $K3$ surface with complex structure I . Following [JK21, Example 7.2], let $\alpha: X \rightarrow X$ be the holomorphic involution swapping the sheets, so that $\text{Fix}(\alpha) = C$ and $\alpha^*\Omega_X = -\Omega_X$ for the holomorphic 2-form Ω_X of X , and let $\beta: X \rightarrow X$ be the lift of complex conjugation $[z] \mapsto [\bar{z}]$ with $\text{Fix}(\beta) = \varpi^{-1}(\mathbb{RP}^2) \cong S^2$; the other lift $\alpha\beta$ acts freely. Note that C has no real points since $F > 0$ on $\mathbb{R}^3 \setminus \{0\}$, so that $\text{Fix}(\beta) \rightarrow \mathbb{RP}^2$ is an unbranched double cover. Let h be the unique Ricci-flat Kähler metric on (X, I) in the Kähler class $\varpi^*[\omega_{FS}]$ (Yau), with Kähler form ω^I , and let ω^J, ω^K be defined by $\Omega_X = \omega^J + i\omega^K$ with Ω_X normalised so that $(X, h, \omega^I, \omega^J, \omega^K)$ is hyperkähler and $\beta^*\Omega_X = \overline{\Omega_X}$. Since α and β preserve, respectively reverse, the Kähler class, uniqueness of h gives, as in [JK21, Example 7.2],

$$\alpha^*(\omega^I, \omega^J, \omega^K) = (\omega^I, -\omega^J, -\omega^K), \quad \beta^*(\omega^I, \omega^J, \omega^K) = (-\omega^I, \omega^J, -\omega^K). \quad (4.1)$$

Let $T^3 = \mathbb{R}^3/\mathbb{Z}^3$ with coordinates (x_1, x_2, x_3) and let $\Gamma = \langle \alpha, \beta \rangle \cong \mathbb{Z}_2^2$ act on T^3 by [JK21, (7.1)],

$$\alpha: (x_1, x_2, x_3) \mapsto (x_1, -x_2, -x_3), \quad \beta: (x_1, x_2, x_3) \mapsto (-x_1, x_2, \tfrac{1}{2} - x_3),$$

and diagonally on $T^3 \times X$. By (1.6) and (4.1), Γ preserves the torsion-free G_2 -structure (φ, g) of $T^3 \times X$, and $M' := (T^3 \times X)/\Gamma$ is a compact torsion-free G_2 -orbifold. As computed in [JK21, Example 7.2], its singular set L has four components:

- $L_1, L_2 \cong S^1 \times C$, the images of $S^1 \times \{(x_2, x_3)\} \times C$ with $(x_2, x_3) \in \{0, \frac{1}{2}\}^2$, fixed by α and exchanged in pairs by β ;
- $L_3, L_4 \cong S^1 \times S^2$, the images of $\{x_1\} \times S^1 \times \{x_3\} \times \text{Fix}(\beta)$ with $x_1 \in \{0, \frac{1}{2}\}$, $x_3 \in \{\frac{1}{4}, \frac{3}{4}\}$, fixed by β and exchanged in pairs by α ;

in each case the metric $g|_{L_i}$ is a product with the circle of length 1, and $\lambda := dx_1$ on L_1, L_2 and $\lambda := dx_2$ on L_3, L_4 is closed, coclosed and nowhere vanishing [JK21, §7.1 (B)]. The resolution N satisfies $b^2(N) = 4$, $b^3(N) = 67$, is simply connected and carries torsion-free G_2 -structures with holonomy G_2 [JK21, Example 7.2]; we equip N with the torsion-free G_2 -structures φ_t of Theorem 2.12.

4.2. Hypothesis (a). Consider L_1 , the image of $S^1 \times \{(0, 0)\} \times C$. Since β maps a small neighbourhood of $S^1 \times \{(0, 0)\} \times C$ in $T^3 \times X$ to a disjoint set, a neighbourhood of L_1 in M' is isometric to a neighbourhood of the image of $S^1 \times \{(0, 0)\} \times C$ in $(T^3 \times X)/\langle \alpha \rangle = S^1 \times W$, where

$$W := (T^2 \times X)/\langle \alpha \rangle, \quad T^2 := \mathbb{R}^2/\mathbb{Z}^2 \ni (x_2, x_3), \quad (4.2)$$

because α acts trivially on the x_1 -circle. Let (ω, Ω) be the torsion-free $SU(3)$ -structure on $T^2 \times X$ induced by φ and the parallel unit vector field ∂_{x_1} , so that $\varphi = dx_1 \wedge \omega + \text{Re } \Omega$ as in [JK21, (2.24)]; explicitly, by (1.6),

$$\omega = \iota_{\partial_{x_1}} \varphi = dx_2 \wedge dx_3 - \omega^I, \quad \text{Re } \Omega = \varphi - dx_1 \wedge \omega = -dx_2 \wedge \omega^J - dx_3 \wedge \omega^K,$$

and its complex structure is $\partial_{x_1} \times (\cdot) = (J_{T^2}, -I)$, where $J_{T^2} \partial_{x_2} = \partial_{x_3}$. By (4.1), (ω, Ω) is α -invariant and descends to W . The singular set of W consists of the four complex curves $\{(x_2, x_3)\} \times C$, $(x_2, x_3) \in \{0, \frac{1}{2}\}^2$, with transverse model $\mathbb{C}^2/\{\pm 1\}$, since α acts by -1 on the normal bundle $T_{(x_2, x_3)} T^2 \oplus N_{C/X}$. Hence (a) holds for L_1 with $\ell = 1$, $\Sigma = \{(0, 0)\} \times C \cong C$ and $\lambda = dx_1$; similarly for L_2 .

For L_3 , the image of $\{0\} \times S^1 \times \{\frac{1}{4}\} \times \text{Fix}(\beta)$, the same argument with α and β exchanged gives a neighbourhood isometric to $S^1 \times U_{W'}$, where the circle is the x_2 -circle and

$$W' := (T^2 \times X)/\langle \beta \rangle, \quad T^2 := \mathbb{R}^2/\mathbb{Z}^2 \ni (x_1, x_3), \quad \beta(x_1, x_3) = (-x_1, \tfrac{1}{2} - x_3), \quad (4.3)$$

and with the torsion-free $SU(3)$ -structure (ω', Ω') on $T^2 \times X$ induced by φ and ∂_{x_2} , i.e. $\varphi = dx_2 \wedge \omega' + \text{Re } \Omega'$ with

$$\omega' = \iota_{\partial_{x_2}} \varphi = dx_3 \wedge dx_1 - \omega'^J, \quad \text{Re } \Omega' = -dx_1 \wedge \omega'^I - dx_3 \wedge \omega'^K,$$

whose complex structure is $\partial_{x_2} \times (\cdot) = (J'_{T^2}, -J)$ with $J'_{T^2} \partial_{x_3} = \partial_{x_1}$; it is β -invariant by (4.1). Since β preserves ω^J and h , it is holomorphic for $-J$, and it acts by -1 on the $(-J)$ -holomorphic 2-form $\omega^K - i\omega^I$ by (4.1); hence $\text{Fix}(\beta) \cong S^2$ is a complex curve in $(X, -J)$, and the singular set of W' consists of four copies of it, with transverse model $\mathbb{C}^2/\{\pm 1\}$. Thus (a) holds for L_3 and L_4 with $\ell = 1$, $\Sigma' = \text{Fix}(\beta) \cong S^2$ and $\lambda = dx_2$.

4.3. Symmetries. Let $\mathcal{G} \subset \text{PGL}(3, \mathbb{R})$ be the group of signed permutation matrices modulo ± 1 ; it has order 24 and is isomorphic to the rotation group of the cube. Every $g \in \mathcal{G}$ preserves F , hence acts on $\mathbb{CP}^2$ preserving C and commuting with complex conjugation, and it lifts to exactly two holomorphic automorphisms of X , which differ by α and commute with α and β . Both lifts preserve the Kähler class $\varpi^*[\omega_{FS}]$, hence h and ω^I , and they act on Ω_X by constants c and $-c$ of modulus one. Since a lift commutes with β and $\beta^* \Omega_X = \overline{\Omega_X}$, we get

$c\overline{\Omega_X} = \overline{c\Omega_X}$, so $c = \pm 1$. We denote by $\tilde{g}$ the lift with $\tilde{g}^*\Omega_X = \Omega_X$, which therefore preserves $\omega^I, \omega^J, \omega^K$ and h , and we set

$$k_g := \text{id}_{T^3} \times \tilde{g} \in \text{Diff}(T^3 \times X). \quad (4.4)$$

By (1.6), k_g preserves (φ, g) ; it commutes with Γ , hence descends to an automorphism of (M', φ, g) , which preserves each component L_i and λ since it acts trivially on T^3 . Under the identifications of the previous subsection, $K_g := \langle k_g \rangle$ acts on $S^1 \times U_W$ and $S^1 \times U_{W'}$ by $\text{id} \times (\text{id}_{T^2} \times \tilde{g})$, as required in (a), and on $\Sigma \cong C$ by $g|_C$ (both lifts of g agree on $C = \text{Fix}(\alpha)$), and on $\Sigma' = \text{Fix}(\beta)$ by $\tilde{g}|_{\text{Fix}(\beta)}$.

4.4. Fixed points. For $j \in \{0, 1, 2\}$ let $s_j \in \mathcal{G}$ be the sign change of z_j , and for $i < j$ and $\varepsilon \in \{\pm 1\}$ let $\tau_{ij}^\varepsilon \in \mathcal{G}$ be the involution which exchanges z_i and z_j , and multiplies the third coordinate by ε .

Lemma 4.5. *Let $c \in C$ be one of the 54 points of*

$$\bigcup_j C \cap \{z_j = 0\} \quad \cup \quad \bigcup_{i < j, \varepsilon} C \cap \{z_i = \varepsilon z_j\},$$

and let $g \in \{s_j, \tau_{ij}^\varepsilon\}$ be the corresponding involution. Then g fixes c and $\text{dg}_c = -1$ on $T_c C$.

Proof. The fixed point set of s_j in $\mathbb{CP}^2$ is the line $\{z_j = 0\}$ together with the point $[e_j]$; that of τ_{ij}^+ (respectively τ_{ij}^-) is the line $\{z_i = z_j\}$ (respectively $\{z_i = -z_j\}$) together with an isolated point. The isolated points do not lie on C , and each of the lines meets C in six distinct points: for instance $\{z_0 = z_1\} \cap C = \{[a, a, b] : 2a^6 + b^6 = 0\}$. By Bézout the intersections are transverse. At such a point c , dg_c has eigenvalues $+1$ on the tangent line of the fixed line and -1 on a complementary line; $T_c C$ is dg_c -invariant and transverse to the fixed line, so $\text{dg}_c = -1$ on $T_c C$. Finally, the nine lines $\{z_j = 0\}$, $\{z_i = \varepsilon z_j\}$ meet pairwise only in points all of whose coordinates lie in $\{0, \pm 1\}$, and F takes the values $1, 2, 3$ at such points; so no two of the lines meet on C , and the 54 points are distinct. $\square$

Lemma 4.6. *Let $\ell \in \mathbb{RP}^2$ be one of the 13 points $[e_j]$, $[e_i \pm e_j]$, $[1, \pm 1, \pm 1]$, and let $p, p' \in \text{Fix}(\beta)$ be its two preimages under ϖ . Then there is $g \in \mathcal{G}$ such that $\tilde{g}$ fixes p and p' , and $\text{d}\tilde{g}_p, \text{d}\tilde{g}_{p'}$ act on $T_p \text{Fix}(\beta), T_{p'} \text{Fix}(\beta)$ as rotations by a non-zero angle.*

Proof. The group $\mathcal{G}$ acts on $\mathbb{RP}^2 = S^2/\{\pm 1\}$ as the rotation group of the cube, and the listed points are the images of the vertices, edge midpoints and face centres of the cube; each of them is an isolated fixed point of an element $g \in \mathcal{G}$ (s_j for $[e_j]$, $\tau_{ij}^\mp$ for $[e_i \pm e_j]$, and a signed cyclic permutation of order three for $[1, \pm 1, \pm 1]$), with dg_ℓ a rotation of $T_\ell \mathbb{RP}^2$ by π or by $\pm 2\pi/3$. In particular the complex-linear extension of dg_ℓ to $T_\ell \mathbb{CP}^2 = T_\ell \mathbb{RP}^2 \otimes \mathbb{C}$ has determinant one. Since $\ell \notin C$, ϖ is a local biholomorphism near p ; one of the two lifts of g fixes p (the other maps p to $\alpha(p) = p'$), and its derivative at p is identified with dg_ℓ , so it preserves Ω_X at p ; as the lifts act on Ω_X by constants, this lift is $\tilde{g}$. The same argument applies to p' . Finally $\tilde{g}$ commutes with β , hence preserves $\text{Fix}(\beta)$, and its derivative at p is the rotation dg_ℓ of $T_p \text{Fix}(\beta) \cong T_\ell \mathbb{RP}^2$. $\square$

4.5. Proof of Corollary 1.3. Let $L_0 \in \{L_1, L_2\}$ and let $c \in C$ be one of the 54 points of Lemma 4.5, with involution g . Then hypotheses (a) and (b) of Theorem 1.1 hold for $K = K_g$, by Section 4.3 and Lemma 4.5. Similarly, for $L_0 \in \{L_3, L_4\}$ and p one of the 26 points of Lemma 4.6, they hold for $K = K_g$ with g as in Lemma 4.6. Theorem 1.1 yields the $2 \cdot 54 + 2 \cdot 26 = 160$ associative submanifolds, which are pairwise distinct for small t : P_t lies within g_t^P -distance $Ct^{1+\theta}$ of Q_c (see the remark after Theorem 1.1), the projection $\sigma: (P^R, g_t^P) \rightarrow (L, g|_L)$ does not increase distances by (2.4) and maps Q_c onto the circle $S^1 \times \{c\}$, and the circles $S^1 \times \{c\}$ for the 54 (respectively 26) points c are at positive distance from each other in L . Their volumes are $\pi t^2(1 + O(t^\theta))$ because $\ell = 1$. Those belonging to the same component L_i are homologous to each other, since $[P_t] = [Q_c]$ and c varies in the connected surface C , respectively $\text{Fix}(\beta)$. They are homologically non-trivial: Z is trivial

here, so $Q \cong L \times S^2$, and $[Q_c]$ is the image of $[S^1 \times \{c\}] \otimes [S^2] \in H_1(L_i) \otimes H_2(S^2) \subset H_3(Q)$, which is non-zero because $[S^1 \times \{c\}] \neq 0$ in $H_1(L_i)$; by [JK21, proof of Proposition 6.1] the summand $H^{k-2}(L)$ of $H^k(N)$ restricts isomorphically onto the summand $H^{k-2}(L) \otimes H^2(S^2)$ of $H^k(Q)$, so that $H_1(L) \otimes H_2(S^2) \rightarrow H_3(N)$ is injective. $\square$

Remark 4.7. The count of Corollary 1.3 is not expected to be sharp, although the subgroups of $\{k_g : g \in \mathcal{G}\}$ give nothing more: the 54 points of Lemma 4.5 are precisely the points of C with non-trivial stabiliser in $\mathcal{G}$ (the elements of order 3 have no fixed points on C , and the fixed points on C of the elements of order 4 lie on the lines $\{z_j = 0\}$), and at the points of $\mathbb{RP}^2$ lying on the fixed line of an involution hypothesis (b) fails. Further associative submanifolds may be obtained from symmetries of M' which act non-trivially on the circle factor in (1.2) but preserve the circles $(\mathbb{R}/\ell\mathbb{Z}) \times \{c\}$, such as $\rho \times \tilde{g}$ with $\rho(x_1, x_2, x_3) = (-x_1, x_2, x_3)$ (this requires a modification of Lemma 3.10), from other K3 surfaces with a large group of symmetries commuting with α and β , and from the examples [JK21, Example 7.3] and [JK21, §7.2], which can be treated in the same way as long as hypothesis (a) holds. Hypothesis (a) excludes components on which λ is twisted along the circle direction, such as the components $(S^1 \times C)/\mathbb{Z}_2$ of [JK21, Example 7.3]; we do not pursue this here.

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